

A Complete Critical Boundary-Orbit Atom and the Double-Alias Compensation Demand

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Abstract

At the physical first alias, RH-338 extracted $2k - 1$ marked points of the primitive boundary orbit from the frozen far cell. We complete that extraction across the full RH-334 raw partition. Let

$$\Gamma_k = \{|f^j(p_k)| : 0 \leq j < 2k\}, \quad G_k = \frac{r_H^{-2k}}{1 + |M_k|}.$$

The folded orbit has exactly $2k$ distinct points, every point has multiplier M_k , and finite-set localization has zero noisy trace. Thus the complete signed raw atom is $-F_k^{\text{orb}}$, where $F_k^{\text{orb}} = 2kG_k$.

The sole point not certified far by RH-338 is $\xi_k = h(p_k) < b$. With $\epsilon_k = \mathbf{1}_{\{\xi_k \in J^-\}}$, the eventual orbit counts in (J^-, J^+, F) are exactly $(\epsilon_k, 0, 2k - \epsilon_k)$. Removing the full orbit before evaluating the deterministic rest gives

$$q_{\sigma,k,2k} = \mathcal{T}_k^{\text{rest}} + \mathcal{P}_{\sigma,2k} - \mathcal{A}_{k,2k} - F_k^{\text{orb}}.$$

Since the direct coefficient is $p = q - d$, critical direct closure requires

$$\mathcal{T}_k^{\text{rest}} + \mathcal{P}_{\sigma,2k} - d_{\sigma,k,2k} = \mathcal{A}_{k,2k} + F_k^{\text{orb}} + o(H_k).$$

The missing point is not target-negligible. If $D_k^{\text{orb}} = (2k - 1)G_k$ is the RH-338 atom, then $F_k^{\text{orb}} - D_k^{\text{orb}} = G_k$ and

$$\frac{G_k}{H_k} = \frac{(\beta R)^{2k}}{C_M k} \{1 + o(1)\} \longrightarrow \infty.$$

Moreover $F_k^{\text{orb}}/\mathcal{A}_{k,2k} \rightarrow 1$, so the required positive side is asymptotically twice the alias packet. This is an exact physical raw-trace decomposition and a necessary compensation law, not a compensation theorem. The orbit-free rest, critical verdict, strict prefix, determinant gluing, and Gates A–E remain open. No Riemann-hypothesis conclusion follows.

1 Source-locked orbit and observation type

Let $u_c \in (1, 2)$ be the real root of

$$u_c^3 - 2u_c^2 + 2u_c - 2 = 0, \tag{1}$$

and put

$$f(x) = 1 - u_c x^2, \quad r = u_c - 1, \quad \lambda = 2u_c r > 1, \quad b = u_c^{-1/2}. \tag{2}$$

The two-step component map is $S = f^2$. Its inverse branch used by the boundary word is

$$h(x) = \sqrt{\frac{1 - \sqrt{(1-x)/u_c}}{u_c}}, \quad h(1) = b. \tag{3}$$

Let $p_k = p_{2k}$ be the primitive point coded by $CA(CB)^{k-1}$. RH-17 proves that the S -orbit is the ordered chain

$$p_k, h^{k-1}(p_k), h^{k-2}(p_k), \dots, h(p_k), \quad (4)$$

and that its physical multiplier obeys

$$M_k = (f^{2k})'(p_k) = (S^k)'(p_k) = -C_M \lambda^k \{1 + o(1)\}, \quad C_M > 0. \quad (5)$$

Primitivity gives $2k$ distinct signed marked points [2, 5].

Work on the exact physical clock and Hardy normalization

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad r_H = \frac{17}{20}, \quad R = \frac{7}{5}, \quad H_k = kR^{-2k}. \quad (6)$$

For fixed $A > 0$, freeze the corrected RH-334 partition

$$J^- = [0, 1] \cap [b - A\sqrt{\sigma}, b), \quad (7)$$

$$J^+ = [0, 1] \cap [b, b + A\sqrt{\sigma}], \quad (8)$$

$$F = [0, 1] \setminus (J^- \cup J^+). \quad (9)$$

For measurable $J \subset [0, 1]$ let

$$L_{\sigma,n}(J) = \text{Tr}(M_J K_\sigma^n), \quad P_n^{\text{abs}}(J) = \sum_{\substack{f^n(x)=x \\ |x| \in J}} \frac{1}{|1 - (f^n)'(x)|}. \quad (10)$$

RH-334 proves that absolute value is a multiplier-preserving bijection from the signed fixed points to the folded fixed points, preserving minimal period and marked-point multiplicity. It also proves the exact raw partition

$$\mathcal{T}_{\sigma,n} := r_H^{-n} \{\text{Tr } K_\sigma^n - P_n\} = \mathcal{B}_{\sigma,k,n} + \mathcal{S}_{\sigma,k,n} + \mathcal{R}_{\sigma,k,n}, \quad (11)$$

where each slot is the localized noisy trace minus the localized deterministic flat trace on the corresponding frozen cell [7]. These are cyclic trace observations, not forward probabilities.

2 Complete cellwise orbit extraction

Define the complete folded marked orbit and its critical member by

$$\Gamma_k = \{|f^j(p_k)| : 0 \leq j < 2k\}, \quad \xi_k = |f^{2k-2}(p_k)| = h(p_k) < b. \quad (12)$$

RH-338 proves that the other $2k - 1$ marked points form a set $\Omega_k \subset F$ eventually. The location of ξ_k is left open there because its distance from b is on the window scale [4].

Put

$$G_k = \frac{r_H^{-2k}}{1 + |M_k|}, \quad \epsilon_k = \mathbf{1}_{\{\xi_k \in J^-\}}. \quad (13)$$

Theorem 2.1 (Complete cellwise boundary-orbit atom). *For all sufficiently large k , Γ_k consists of $2k$ distinct folded marked points with common flat-trace weight $(1 + |M_k|)^{-1}$. Its counts in (J^-, J^+, F) are*

$$(\epsilon_k, 0, 2k - \epsilon_k). \quad (14)$$

Define orbit-free rest slots by applying the same localized raw ledger to $J^- \setminus \Gamma_k$, $J^+ \setminus \Gamma_k$, and $F \setminus \Gamma_k$. Then exactly

$$\mathcal{B}_{\sigma,k,2k} = \mathcal{B}_k^{\text{rest}} - \epsilon_k G_k, \quad (15)$$

$$\mathcal{S}_{\sigma,k,2k} = \mathcal{S}_k^{\text{rest}}, \quad (16)$$

$$\mathcal{R}_{\sigma,k,2k} = \mathcal{R}_k^{\text{rest}} - (2k - \epsilon_k) G_k. \quad (17)$$

Consequently, with

$$\mathcal{T}_k^{\text{rest}} = \mathcal{B}_k^{\text{rest}} + \mathcal{S}_k^{\text{rest}} + \mathcal{R}_k^{\text{rest}}, \quad F_k^{\text{orb}} = 2kG_k, \quad (18)$$

the full raw trace has the partition-independent decomposition

$$\boxed{\mathcal{T}_{\sigma,2k} = \mathcal{T}_k^{\text{rest}} - F_k^{\text{orb}}}. \quad (19)$$

Proof. The folding bijection and primitivity give the cardinality and common multiplier. Equation (5) is negative eventually, so each physical weight is $(1 + |M_k|)^{-1}$. The $2k - 1$ points of Ω_k lie in F , while $\xi_k < b$ lies in either J^- or F and never in J^+ . This proves (14).

A finite set has zero multiplication operator on $L^2[0, 1]$. Therefore its localized noisy trace vanishes exactly. Removing a marked point changes a raw cell ledger only by deleting its positive deterministic weight, which proves (15)–(17). Summation cancels the allocation ϵ_k and gives (19). $\square$

Remark 2.2 (Orbit-free does not mean empty or diffuse). The rest still contains the noisy trace over a full-measure set and every other deterministic periodic point. No sign, localization estimate, or smallness claim is attached to it. The analytic subpartition exposes an exact constituent of the already-frozen physical observation; it does not replace that observation by a fitted window.

3 The typed critical coefficient

Retain the RH-334 parity and alias packets

$$\mathcal{P}_{\sigma,n} = r_H^{-n} \{(-1)^n - \lambda_-(\sigma)^n\}, \quad (20)$$

$$\mathcal{A}_{k,n} = s_{k,n} - p_n^{\text{pole}}. \quad (21)$$

At $n = 2k$,

$$\mathcal{A}_{k,2k} = (2k - 2)\beta_k^{2k} + 2\beta_k^{2k}, \quad \beta_k^{2k} = \frac{r_H^{-2k}}{|M_k|}, \quad \beta = \frac{1}{r_H \sqrt{\lambda}}. \quad (22)$$

The Hardy full-trace constituent and direct modulus-complement coefficient obey

$$q_{\sigma,k,n} = \mathcal{T}_{\sigma,n} + \mathcal{P}_{\sigma,n} - \mathcal{A}_{k,n}, \quad p_{\sigma,k,n} = q_{\sigma,k,n} - d_{\sigma,k,n}, \quad (23)$$

where $d_{\sigma,k,n} = h_{\sigma,n} - s_{k,n}$ is the separate noisy-head/counterloop defect [7].

Theorem 3.1 (Complete critical coefficient and compensation demand). *At the physical critical order,*

$$q_{\sigma,k,2k} = \mathcal{T}_k^{\text{rest}} + \mathcal{P}_{\sigma,2k} - \mathcal{A}_{k,2k} - F_k^{\text{orb}}, \quad (24)$$

$$p_{\sigma,k,2k} = \mathcal{T}_k^{\text{rest}} + \mathcal{P}_{\sigma,2k} - d_{\sigma,k,2k} - \mathcal{A}_{k,2k} - F_k^{\text{orb}}. \quad (25)$$

Consequently

$$p_{\sigma,k,2k} = o(H_k) \iff \mathcal{T}_k^{\text{rest}} + \mathcal{P}_{\sigma,2k} - d_{\sigma,k,2k} = \mathcal{A}_{k,2k} + F_k^{\text{orb}} + o(H_k). \quad (26)$$

In particular, vanishing of the direct weighted prefix at the strict cut $2 \leq n < 4k$ implies (26).

Proof. Substitute (19) into (23); this gives (24) and (25). Rearrangement proves the equivalence. The critical summand in the nonnegative direct prefix is

$$\frac{|p_{\sigma,k,2k}| R^{2k}}{2k} = \frac{|p_{\sigma,k,2k}|}{2H_k}.$$

Thus prefix vanishing forces $p_{\sigma,k,2k} = o(H_k)$. $\square$

This theorem expands the quantity used by RH-340. If $D_k^{\text{orb}} = (2k - 1)G_k$ and $C_k^0 = q_{\sigma,k,2k} + D_k^{\text{orb}}$, then exactly

$$C_k^0 = \mathcal{T}_k^{\text{rest}} + \mathcal{P}_{\sigma,2k} - \mathcal{A}_{k,2k} - G_k. \quad (27)$$

Hence the RH-340 law $C_k^0 - d = D_k^{\text{orb}} + o(H_k)$ is algebraically equivalent to (26); the advance is the physical decomposition of its previously unexpanded combined complement [9].

4 The omitted marked point is super-target

Theorem 4.1 (Complete-orbit and missing-point scales). *As $k \rightarrow \infty$,*

$$F_k^{\text{orb}} = \frac{2k}{C_M} \beta^{2k} \{1 + o(1)\}, \quad (28)$$

$$\frac{F_k^{\text{orb}}}{\mathcal{A}_{k,2k}} \longrightarrow 1, \quad (29)$$

$$\frac{F_k^{\text{orb}}}{H_k} = \frac{2}{C_M} (\beta R)^{2k} \{1 + o(1)\} \longrightarrow \infty. \quad (30)$$

Moreover

$$\frac{F_k^{\text{orb}}}{D_k^{\text{orb}}} = \frac{2k}{2k - 1}, \quad F_k^{\text{orb}} - D_k^{\text{orb}} = G_k, \quad (31)$$

but

$$\frac{G_k}{H_k} = \frac{1}{C_M k} (\beta R)^{2k} \{1 + o(1)\} \longrightarrow \infty. \quad (32)$$

Finally,

$$\mathcal{A}_{k,2k} + F_k^{\text{orb}} = \frac{4k}{C_M} \beta^{2k} \{1 + o(1)\}, \quad (33)$$

$$\frac{\mathcal{A}_{k,2k} + F_k^{\text{orb}}}{\mathcal{A}_{k,2k}} \longrightarrow 2. \quad (34)$$

Proof. Equation (5) gives

$$G_k = \frac{1}{C_M} \beta^{2k} \{1 + o(1)\}.$$

Multiplication by $2k$ proves (28). RH-338 gives the same leading asymptotic for (22), proving (29). Division by H_k gives (30). The exact RH-336 certificate $\beta R > 1$ makes the limit infinite [4, 8].

Equations (31) are immediate from the exact marked-point counts. Dividing the asymptotic for G_k by H_k proves (32); an exponential with base $\beta R > 1$ dominates k . Adding the two equal leading asymptotics proves (33)–(34). $\square$

Thus the relative statement $F_k^{\text{orb}} = D_k^{\text{orb}} \{1 + o(1)\}$ is true, but the target-scale replacement $F_k^{\text{orb}} = D_k^{\text{orb}} + o(H_k)$ is false. The exact finite- k normalization also gives

$$\mathcal{A}_{k,2k} - F_k^{\text{orb}} = 2(\beta^{2k} - \beta_k^{2k}) + \frac{2k\beta_k^{2k}}{1 + |M_k|}. \quad (35)$$

The repository does not lock the sign of this expression, and none is used.

Relative to the positive demand in (26), target closure requires precision

$$o\left(\frac{H_k}{\mathcal{A}_{k,2k} + F_k^{\text{orb}}}\right) = o\left((\beta R)^{-2k}\right). \quad (36)$$

This remains a signed aggregate requirement. Taking separate absolute values of the alias, orbit, parity, rest, and head terms cannot demonstrate the cancellation.

5 Phase-resolved cell allocation

The aggregate atom is independent of the unresolved cell allocation, but the latter can be stated sharply. Define

$$q_{b,\sigma,k} = \frac{b - h(p_k)}{\sqrt{\sigma}}, \quad \eta_\sigma = k - \frac{\log(1/\sigma)}{2 \log \lambda}. \quad (37)$$

The half-open endpoint convention gives the exact finite relation

$$\epsilon_k = \mathbf{1}_{\{q_{b,\sigma,k} \leq A\}}. \quad (38)$$

RH-327 proves, along $\eta_\sigma \rightarrow \eta$,

$$q_{b,\sigma,k} \longrightarrow Q(\eta) := \frac{\sqrt{C_b} \lambda^{-\eta}}{2u_c}. \quad (39)$$

Therefore ϵ_k is eventually one if $Q(\eta) < A$ and eventually zero if $Q(\eta) > A$. At $Q(\eta) = A$, the available $o(1)$ expansion does not decide stabilization. Exact finite equality belongs to J^- because its left endpoint is included. This phase sensitivity is why F_k^{orb} is a complete raw-partition atom, not always a far atom [6].

6 Executable audit and claim boundary

The artifact reconstructs the boundary orbit at $k = 2, 4, 8, 16, 32$, checks all $2k$ folded points and their cell counts, evaluates the exact missing-point and complete-orbit identities, and reproduces the approach to the scale ratios. A rational fixture independently verifies the typed equations (24)–(25). These finite rows are formula checks only, never all-order or physical-convergence evidence.

The exact new edge is limited to the complete physical orbit subledger, its cellwise raw decomposition, the target-supercritical missing point, and the necessary compensation law. No theorem estimates $\mathcal{T}_k^{\text{rest}} - d_{\sigma,k,2k}$ at moving order. Hence critical signed compensation and critical nonclosure both remain `NOT.TESTABLE/open`. No conclusion follows for the actual D_{4k} head budget, the lower sideband, the remaining off-alias background, or the direct strict prefix.

RH-343's finite normal models are not substituted for this physical raw trace [3]. RH-288 remains inactive, and the ten-layer frontier of RH-341 is not promoted [1]. Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

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